SMOLDER: forecasting next-three-day wildfire risk for continental Australia at 1 km with a dual-branch ConvLSTM (2015–2020)

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Plain-language summary

Fire agencies need to know where fire is likely in the coming days. SMOLDER is a computer model trained on satellite and weather maps of Australia from 2015 to 2019 and tested on 2020. Each day it gives every 1 km pixel of land a risk score for fire in the next three days. On an average day of 2020, the 0.07 % of the land with the highest scores (about 5400 km², a little more than twice the size of the Australian Capital Territory) contained 30 % of the fire of the next three days, and about one in seven of the flagged pixels burned. A simple rule of thumb, 'expect fire next to recent fire', needed twice the area to catch slightly less fire. The model is therefore good at telling where burning fires will spread or flare up again. It almost never foresees fires that start far from existing fires, which made up two fifths of the fire in 2020. Foreseeing those needs other information, such as lightning.

Abstract. Whether a wildfire starts and spreads depends on the amount and dryness of the fuel, which change over months, and on the fire weather, which changes over days. This study presents SMOLDER (Slow-Memory Operator with Linked Dual branches for Estimating fire Risk), a deep learning model with one branch for each time scale. For every 1 km land pixel of continental Australia, it forecasts whether a satellite will detect fire within the next three days. One branch reads 144 days of leaf area index, soil moisture and precipitation, the other 14 days of weather, vegetation greenness and recent fire. All inputs were checked for information from after the day of the forecast. Trained on 2015 to 2018, tuned on 2019 and evaluated on 2020, the model is compared with a persistence baseline that ranks pixels by their distance to recent fire. In 2020 its area under the precision-recall curve was 0.1206, 3.3 times that of the baseline (0.0361) and about 380 times the base rate of 0.0317 %, and its daily value was higher on 348 of 350 days. With one score threshold chosen on 2019, SMOLDER flags 0.071 % of the land per day and catches 30.5 % of the fire, where the baseline flags 0.140 % and catches 27.8 %. Of the fire it catches, 91.1 % lies within 3 km of recent fire, and of the 41.7 % of the fire that starts more than 10 km away it catches 0.6 %. SMOLDER therefore forecasts where existing fires spread and flare up again, not where new fires start. In practice, such a ranking can guide surveillance and crews around burning fires, whereas new ignitions still call for fire danger ratings and lightning data. Code, weights and evaluation data are openly available.

1 Introduction

Fire activity can be described as the result of four 'switches': enough fuel, fuel dry enough to burn, weather that lets fire spread, and an ignition (Bradstock, 2010). All four have to be on at the same time (Nolan et al., 2016). Fuel amount and dryness change slowly. The moisture of living plants depends on soil moisture, leaf area and vegetation type and changes over weeks to months (Nolan et al., 2016; McNorton and Di Giuseppe, 2024), whereas dead leaves, grass and twigs can dry out within a week in response to the weather (Matthews, 2014; Nolan et al., 2016). Fire becomes likely once fuel moisture falls below critical thresholds (Nolan et al., 2016), and how likely fire is at a given fuel moisture differs between vegetation types (McNorton and Di Giuseppe, 2024). Satellites and weather reanalyses, consistent reconstructions of past weather from observations and models, now describe both time scales over large areas (Yebra et al., 2013; Su et al., 2025), which makes data-driven forecasts of fire possible.

Deep learning models, which learn such relationships from large amounts of example data instead of following hand-written rules, are increasingly used for this task. Many of them receive the fire that is already burning as an input (e.g. Huot et al., 2022; Gerard et al., 2023). Such a model can score high by predicting that fire continues where it is. Its skill therefore has to be judged against this 'persistence' forecast and separately for fire at new places. Fire is also extremely rare at the scale of single pixels, so scores depend strongly on how the evaluation is set up, and inputs

derived from satellite products can, unnoticed, contain information from after the day of the forecast, which makes a model look better than it is.

This paper presents SMOLDER (Slow-Memory Operator with Linked Dual branches for Estimating fire Risk), a model that forecasts for every 1 km land pixel of continental Australia whether a satellite will detect fire within the next three days. The emphasis is on a fair evaluation: for the whole continent, at the true rarity of fire, against persistence, and as a function of the distance to existing fire. Three questions are asked: how much better than persistence is the model, which part of the fire can it anticipate, and what does it rely on?

2 Data

2.1 Study region and grid

The study area is continental Australia without Tasmania, and the study period is 2015 to 2020. All datasets (Table 1) were brought to the grid of the SMIPS soil moisture product, 3474 × 4110 pixels of 0.01° (about 1 km, or 1.1 km² on average), of which 6.9 million are land, and stored as yearly data cubes in the .zarr format, which the SeasFire cube also uses (Karasante et al., 2025).

2.2 Input layers and their timing

Fire. Fire comes from the VIIRS 375 m active fire product, which detects fires that burn while the satellite passes overhead, around 01:30 and 13:30 local time (Schroeder et al., 2014). Detections classed as presumed vegetation fires of nominal or high confidence (Giglio, 2024) were aggregated to daily binary (fire or no fire) maps on the 1 km grid. The maps of the three days after a forecast form the target, the quantity to be predicted, and the maps up to the issue day form the fire history. The same detections provide the fire radiative power (FRP), a measure of fire intensity (Giglio, 2024), stored per pixel and day as summed FRP, number of detections and share of night detections, and the fuel age, the number of days since a pixel last burned, capped at 1095 days.

Weather. The weather comes from the BARRA-C2 reanalysis at 4.4 km as daily values, resampled bilinearly to 1 km: the mean 10 m wind speed and its two components, the maximum near-surface air temperature and the vapour pressure deficit (VPD) at that temperature, computed from the BARRA-C2 temperature and humidity. VPD measures the atmospheric demand for water and is a reliable predictor of how dry dead fuel is (Clarke et al., 2022), and wind drives the spread of fire (Cruz and Alexander, 2019).

Vegetation, soil and terrain. The leaf area index (LAI), the leaf area per unit of ground area, describes the structure of the canopy (Yan et al., 2024) and is used to estimate the amount of live foliage fuel (McNorton and Di Giuseppe, 2024). Because HiQ-LAI smooths every 8-day composite with its neighbours in time, with a half-length of three composites (Yan et al., 2024), each day is given the newest composite that started at least 31 days earlier. NDVI, computed from the MOD09A1 surface reflectance (Vermote, 2021), measures how green the vegetation is and is used to estimate the moisture of live vegetation (Chuvieco et al., 2004). Each day is given the newest MOD09A1 composite whose eight days have all passed, so that it is at most seven days old. The SMIPS soil moisture index gives the fraction to which the upper 90 cm of the soil is filled, and ANUClimate 2.0 the daily precipitation. Biomass approximates the fuel load, land cover and climate zones separate fuel types and fire regimes, and terrain steers wind and fire (Sharples et al., 2012). A forecast uses the biomass map of the year before the issue day and the land-cover map of two years before, because each land-cover map is made from three years of data.

2.3 Keeping information from the future out of the inputs

A forecast may only use what is known on the day it is issued. If an input already contains the trace of a fire, for ex-

Table 1. Datasets, their native resolution, the model branch they enter and what they tell the model.

Variable	Dataset	Resolution	Branch	What it tells the model
Active fires, fire radiative power	VIIRS 375 m active fires (Schroeder et al., 2014; Giglio, 2024)	375 m, daily	target, fast	where fire burns, how intensely, time since last fire
Leaf area index (LAI)	HiQ-LAI (Yan et al., 2024)	500 m, 8-day	slow	canopy structure and live foliage
NDVI	MODIS MOD09A1 (Vermote, 2021)	500 m, 8-day	fast	greenness, which falls as grass dries out
Soil moisture	SMIPS (Stenson et al., 2021)	1 km, daily	slow	how dry the landscape is
Precipitation	ANUClimate 2.0 (Hutchinson et al., 2021)	1 km, daily	slow	rain that wets fuel and grows grass
Wind, VPD, air temperature	BARRA-C2 (Su et al., 2024, 2025)	4.4 km, hourly	fast	fire weather: drying of fuel, spread of fire
Biomass	ESA CCI Biomass (Santoro and Cartus, 2025)	100 m, annual	both	amount of fuel
Land cover	Copernicus LC100 (Buchhorn et al., 2020)	100 m, annual	both	type of vegetation
Climate zones	Köppen–Geiger (Beck et al., 2023)	1 km	both	climate and fire regime
Terrain	ETOPO1 (NOAA National Geophysical Data Center, 2009)	1′ (≈1.85 km)	both	slopes steer wind and fire

ample a drop in greenness caused by the fire itself, a model can appear to predict a fire that it has in fact already seen. Such look-ahead, also called leakage, can enter satellite and reanalysis products unnoticed in four ways. Composites are often labelled by their first day but contain later days. Products smoothed over time spread an observation to earlier dates. Gap filling with a fit over the whole year carries later observations back, as in the gap-filled MODIS land surface temperature of Zhang et al. (2022). And static maps summarized over the study period contain the years that are forecast. In addition, no target window may cross the boundary between the training, validation and test years.

Each of these was handled by the timing rules of Sect. 2.2: complete composites only, a delay of 31 days for the smoothed LAI, air temperature from the reanalysis instead of a gap-filled land surface temperature, maps that predate the issue day, and no issue days whose target window crosses the end of a year. The climate zones refer to the period 1991 to 2020, but a climate zone cannot encode a single fire.

Two tests then checked the inputs exactly as the model reads them. First, the fire-derived inputs were rebuilt from the raw daily detections and compared pixel by pixel on 60 fire-active patches (18 million comparisons). They differed on at most 1.2×10^{-6} of the pixels, whereas a fire map shifted by one day differs on 8.3×10^{-4}. Second, all other inputs were examined around 320 new fires of 2019, 40 in each of eight fire regions. For a forecast issued 1 to 3 days before a fire is first detected, the fire lies in the target window, so an input without look-ahead cannot show it. The input at the fire pixel was compared with a ring 15 to 40 pixels around it, relative to 16 days before the detection, and counted as clean if this standardized signal stayed below 3. Every input passed (largest value 2.4). Applied to versions of NDVI and LAI that do contain look-ahead, the same test detects it. It finds strong look-ahead, but it cannot prove the absence of a weak one.

3 Methods

3.1 Forecast task and data split

A forecast is issued for a day D, the issue day. For every land pixel, the model gives a score between 0 and 1 for the event that at least one VIIRS active fire is detected there on day D+1, D+2 or D+3. All inputs refer to day D or earlier. The score ranks pixels by risk and is not a calibrated probability. The years 2015 to 2018 were used for training (1283 issue days), 2019 for validation (349) and 2020 for testing (350). Whole years were held out, because a fire lasts several days and a random split of days would put parts of the same fire into training and test. Everything that had to be chosen, that is the inputs, the training settings, the final model and the thresholds of Sect. 3.2, was fixed on 2019. The test year was evaluated twice, once for a provisional version of the model and once for the final model, and nothing was chosen from either evaluation. The architecture and the loss go back to earlier development in which 2020 had also been consulted, so in this respect 2020 is not fully untouched.

3.2 How skill is measured

Each issue day, the model gives every land pixel a score, and the pixels with the highest scores can be flagged. Each pixel then falls into one of four outcomes: caught fire (flagged and burned), missed fire (burned, not flagged), false alarm (flagged, no fire) and other land. The measures that summarize these outcomes are explained in Box 1. Because fire is so rare, all scores are read relative to the base rate. The main summary measure is the AUC-PR, because it is decided by the top of the ranking, where a user would look. Thresholds are chosen by F_2, which suits a use in which a missed fire costs more than a false alarm.

All measures were computed for the whole continent on every valid issue day (6.9 million land pixels per day), with the national map assembled from overlapping tiles of 384 pixels. AUC-PR and ROC-AUC were computed over all days together, with all fire pixels and a weighted sample of one in 500 of the others, and the AUC-PR also per day. The

Box 1. Key terms

Training, validation, test	The model learns from 2015–2018, all choices are made on 2019, and 2020 is kept aside to measure performance on unseen data, like a final exam.
Look-ahead (leakage)	Information from after the issue day that slips into an input, for example a fire scar. It makes a model look better than it is.
Persistence baseline	The rule of thumb ‘fire is most likely next to recent fire’: pixels are ranked by their distance to recent fire.
Recent fire, new fire	Recent fire is fire detected on the issue day or the two days before. New fire lies more than 3 km from any recent fire.
Base rate	Share of land pixels with fire in the three days after an issue day (0.0317 % in 2020).
Fire caught (recall)	Share of all fire pixels that were flagged.
Precision	Share of flagged pixels that burned. A random selection has a precision equal to the base rate.
AUC-PR (average precision)	Precision averaged over all sizes of the flagged area. It equals the base rate for a random ranking and 1 for a perfect one and is decided by the top of the ranking.
ROC-AUC	Probability that a random fire pixel scores higher than a random pixel without fire (0.5 random, 1 perfect). High for any reasonable ranking, because most land is far from fire.
F_1, F_2	Combine precision P and recall R: $F_\beta = (1+\beta^2)PR/(\beta^2 P+R)$. F_2 weighs missed fire more than false alarms.
ConvLSTM	A neural network that reads a sequence of maps like the frames of a film and keeps a memory for every pixel and its neighbourhood.

forecast is compared with a persistence baseline that ranks every land pixel by its distance to the nearest fire detected on the issue day or the two days before (recent fire). This is what an analyst without a model would do: watch the surroundings of current fires. The baseline is evaluated in the same way, and the share of days on which SMOLDER has the higher daily AUC-PR is given with a 95 % bootstrap interval of the mean daily difference.

Where pixels are flagged, one score threshold is used for all days, so that the flagged area grows and shrinks with the fire situation of each day. It is the threshold with the highest F_2 on the validation year 2019, applied unchanged to 2020. Persistence gets its own threshold in the same way. At this threshold it flags the pixels that burned on the issue day or the two days before and their eight neighbours, that is, all pixels up to 1.4 km from recent fire. Thresholds were not chosen on the training years, because the model fits them better than new data and because their patches hold far more fire than the land does (Sect. 3.4). The validation year is unseen during fitting and is evaluated over the whole continent at the true base rate. To separate fire that continues from fire at new places, the fire caught is also computed by distance to recent fire, and new fire is defined as fire more than 3 km from any recent fire.

3.3 Model inputs

Each sample is a patch of 384 × 384 pixels with two stacks of input maps, one for each branch (Fig. 1a, maps of all inputs in the supplement of the code repository). The slow stack holds 18 consecutive 8-day averages of LAI, soil moisture and precipitation, the newest ending on or before the issue day. The fast stack holds the 14 days up to the issue day: weather (VPD, maximum air temperature, wind speed and direction), NDVI, the fire history and the proximity to recent fire, FRP, the fuel age and the downwind alignment, which is high where the wind blows from a recent fire towards the pixel. Both stacks also contain static layers and the day of the year (full list in Appendix A).

The two time scales follow a lag analysis in which each predictor alone was tested for how well it separates pixels with fire in the next three days from pixels without (ROC-AUC over 2015 to 2019, lags of 0 to 180 days). Leaf area index, NDVI, soil moisture and precipitation did so best at the longest lag tested, about 180 days (leaf area index 0.79 against 0.73 at lag 0), and VPD and air temperature at lags of 1 and 10 days. This partly reflects the seasons: in the tropical north, the annual wet and dry seasons supply both fuel growth and dry fuel every year (Bradstock, 2010). The fire inputs were selected on 2019, and alternatives that were not adopted are listed in Appendix B.

3.4 Model architecture and training

SMOLDER consists of two encoders, one for each branch, a step that combines them and an output layer (Fig. 1b, details in Appendix A). Each encoder is a Convolutional Long Short-Term Memory network, or ConvLSTM (Shi et al., 2015), which reads a sequence of maps step by step, like the frames of a film. For every pixel it keeps a memory (a hidden state of 64 numbers) that it updates at each step from the new inputs and from the memory of the pixel and its 5 × 5 neighbourhood, so that it can learn both how conditions develop over time and how a pixel relates to its surroundings. The slow encoder ends with one summary of the fuel and moisture history per pixel, and the fast encoder gives a summary at each of its 14 steps. A learned transformation of the slow summary is added to each fast sum-

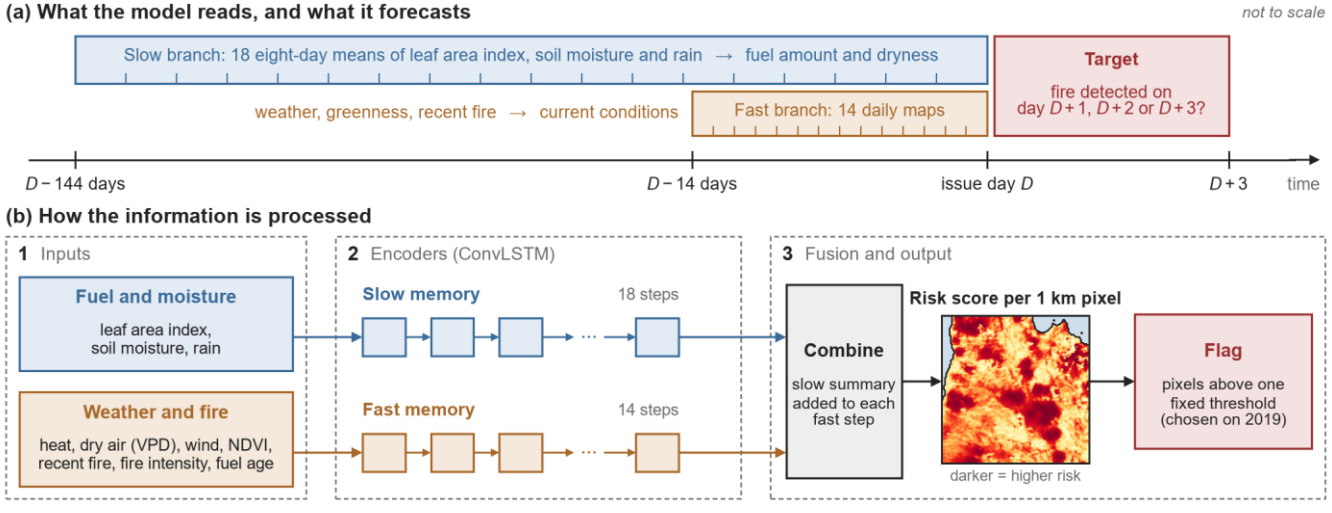


Figure 1. SMOLDER in brief. (a) For an issue day *D*, the slow branch reads 144 days of fuel and moisture information and the fast branch 14 days of weather, greenness and fire. The target is fire detected on days *D*+1 to *D*+3. Terrain, biomass, land cover, climate zone and the day of the year enter both branches. (b) Each branch is a ConvLSTM, which reads its sequence of maps step by step and keeps a memory for every pixel and its neighbourhood. The two are combined into a risk score for every pixel, and the pixels above one fixed threshold are flagged. The architecture is detailed in Appendix A, and maps of all inputs are in the supplement of the code repository.

mary, and a final layer turns the result into a score between 0 and 1. The map of the last fast step is the forecast. The model has 1.19 million adjustable parameters.

Training adjusts these parameters until the scores match the observed fire as well as possible. The model saw 2000 patches from the training years in each of 50 rounds (epochs), half of them drawn to contain fire, because random patches would hardly ever contain any. So that the model does not rely on the fire record alone, in 30 % of the patches fire had to occur where the newest fire map showed none, and in 30 % of the samples all inputs that reveal recent fire were set to 'no fire'. The training objective was a weighted binary cross-entropy over land pixels (as in Huot et al., 2022), in which a missed fire pixel counts more than a false alarm, and the weights of the three best epochs on 2019 were averaged (Appendix A).

3.5 Analysis of the inputs

To see which inputs the forecast relies on, each input was shuffled in turn across the land pixels of a patch, in the same way at every time step, on 1500 fire-active patches of 2020. This permutation importance follows Breiman (2001): an input the model relies on cannot be shuffled without damage to the forecast. Two measures are reported, the share of the flagged pixels that are no longer flagged and the relative fall of the AUC-PR. Independently of the model, the inputs before the issue day were compared between caught fire, missed fire, false alarms and other land of the same patch and day.

4 Results

4.1 Skill compared with persistence

Table 2 gives all scores of both rankings. On the validation year 2019 the AUC-PR of SMOLDER was 0.2091, 3.4 times that of the persistence baseline (0.0609), and its daily AUC-PR was higher on all 349 issue days. In the test year 2020 fire was rarer, with a base rate of 0.0317 % against 0.0543 %. The AUC-PR of SMOLDER was 0.1206, 3.3 times that of the baseline (0.0361), and its daily AUC-PR was higher on 348 of 350 days, with a mean difference of 0.0750 (95 % interval 0.0693 to 0.0807). The daily AUC-PR of both rankings varied strongly through the year. Relative to the base rate the skill hardly changed between the years (385 and 381 times the base rate for SMOLDER, 112 and 114 times for the baseline).

Persistence has a slightly higher ROC-AUC (0.9358 against 0.9302 in 2020), because it ranks all land by distance and so places fire 10 to 25 km from recent fire above all land further away. Within each distance band up to 25 km, SMOLDER orders the land better (ROC-AUC 0.8648 against 0.7016 within 3 km). SMOLDER ranks the top of the list better, and the top is where a user looks: the 0.5 % of the land with the highest score contains 45.9 % of the fire and 20.7 % of the new fire, against 39.3 % and 8.3 % for the baseline (Table 2). The same holds for every flagged area from 0.1 % to 10 % of the land, where the two rankings converge.

Table 2. Measures over all issue days and land pixels for SMOLDER (S) and the persistence baseline (P), on the validation year 2019 and the test year 2020. Fire caught at 0.5 % is the share of the fire of the next three days that lies in the 0.5 % of the land with the highest risk, as a mean over the days.

	2019 (349 days)		2020 (350 days)	
Measure	S	P	S	P
Base rate	0.0543 %		0.0317 %	
AUC-PR	0.2091	0.0609	0.1206	0.0361
Mean daily AUC-PR	0.1808	0.0595	0.1127	0.0377
ROC-AUC	0.9299	0.9355	0.9302	0.9358
AUC-PR, new fire	0.0093	0.0036	0.0058	0.0022
Fire caught, 0.5 % of land	54.1 %	46.2 %	45.9 %	39.3 %
New fire caught, 0.5 %	24.9 %	8.7 %	20.7 %	8.3 %
Days with higher AUC-PR	100 %		99.4 %	

4.2 How much land has to be flagged

With the F_2 threshold of 2019, SMOLDER flags 0.071 % of the land on an average day of 2020 and catches 30.5 % of the fire with a precision of 0.1365 (Table 3), so about one in seven flagged pixels burned. The baseline flags 0.140 % and catches 27.8 % with a precision of 0.0630. SMOLDER thus flags half the area, catches more fire and has 2.2 times the precision. The threshold transferred well between the years: its F_2 on 2020 (0.2445) is practically the best any threshold reaches (0.2447), and it flagged less land in the quieter year (0.071 % against 0.111 %), because the daily flagged area follows the daily amount of fire (rank correlation 0.78).

Figure 2 shows how much land has to be flagged to catch a given share of the fire. To catch half of the 2020 fire with

Table 3. Operating point. The threshold with the highest F_2 was chosen on the validation year 2019 and applied unchanged to 2020. Area: mean share of the land flagged per day. False alarms per fire caught: 6.3, for example, means that for every flagged pixel that burned, 6.3 flagged pixels did not. The 'caught' rows give the share of the fire in each band of distance to recent fire that is caught.

	SMOLDER		Persistence	
	2019	2020	2019	2020
F_2	0.3294	0.2445	0.2214	0.1651
F_1	0.2615	0.1885	0.1473	0.1027
Fire caught (recall)	39.8 %	30.5 %	33.3 %	27.8 %
Precision	0.1946	0.1365	0.0945	0.0630
Area	0.111 %	0.071 %	0.191 %	0.140 %
False alarms per fire caught	4.1	6.3	9.6	14.9
Caught ≤ 3 km from recent fire	78.5 %	72.0 %	72.0 %	72.0 %
Caught 3–10 km away	15.7 %	12.5 %	0.0 %	0.0 %
Caught > 10 km away	0.6 %	0.6 %	0.0 %	0.0 %

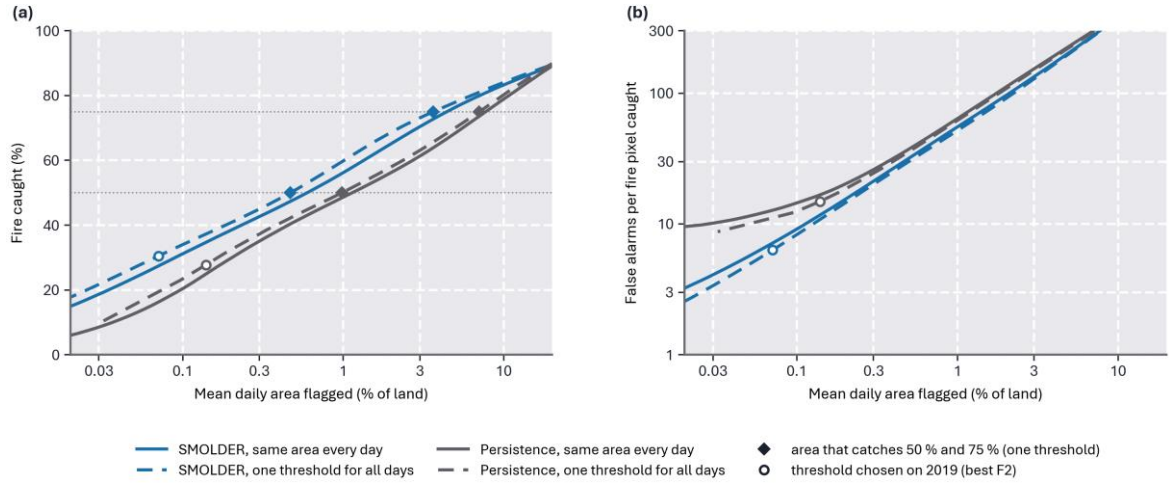


Figure 2. Fire caught and false alarms against the area flagged, test year 2020. (a) Share of the fire of the next three days that lies in the flagged area, against the mean daily share of Australia's land flagged (higher is better). (b) False alarms per fire pixel caught (lower is better). Solid: the same share of land every day. Dashed: one score threshold for all days, so that the flagged area follows the fire danger of the day. Blue: SMOLDER, grey: persistence. Circles: thresholds chosen on 2019 (best F_2). In 2020 they flag 0.071 % (SMOLDER) and 0.140 % (persistence) of the land per day on average. Diamonds: the mean daily area that catches 50 % and 75 % of the fire with one threshold for all days.

one threshold, SMOLDER had to flag 0.468 % of the land per day on average and persistence 0.979 %, and for three quarters 3.632 % and 7.015 %. Catching 90 % or more required a fifth of the land or more for both. From there on persistence needs less area (95 %: 31 % against 45 %), because the last part of the fire lies far from earlier fire, where ranking by distance works better. At such areas no ranking is useful as a warning.

Fewer fires were caught in 2020 than in 2019 (30.5 % against 39.8 %), mainly because more of the 2020 fire lay more than 10 km from recent fire (41.7 % against 32.6 %). Of the decline, 5.7 to 6.2 percentage points come from this shift and 3.2 to 3.7 points from lower catch rates near fire, depending on the order in which the two are separated.

4.3 Skill by distance to recent fire

Most of the skill lies near recent fire (Fig. 3). Persistence catches all fire within 1 km of recent fire and nothing beyond 2 km. SMOLDER catches somewhat less right at the fire (89.6 % of the pixels that burned themselves and 78.8 % of the fire up to 1 km), but also 44.6 % of the fire 2 to 3 km away and 23.9 % at 3 to 5 km, with half the flagged area. Beyond 10 km, where 41.7 % of the 2020 fire lies, neither catches fire (SMOLDER 0.4 % at 10 to 25 km and 0.7 % further away). Of all the fire SMOLDER catches, 91.1 % lies within 3 km of recent fire. Appendix C shows examples of both situations. On 21 November 2020 in northern Queensland SMOLDER caught 75.8 % of 479 fire pixels with 290 false alarms, the baseline 28.2 % with 689. On 6 February 2020 in the south-west of Western Australia, 1065 fire pixels burned without any fire nearby on the days before, and SMOLDER caught 0.7 % of them.

4.4 What the forecast relies on

The three inputs derived from the fire record lead (Fig. 4). When the fire history is shuffled, 83.1 % of the flagged pixels are no longer flagged and the AUC-PR falls by 52.4 %. For the FRP the two values are 52.0 % and 28.9 %, for the fuel age 13.3 % and 7.0 %, and NDVI follows with 15.8 % and 7.1 %. The weather inputs change at most 7.0 % of the flagged pixels and lower the AUC-PR by at most 0.8 %. The model thus relies mainly on where fire burned, how intensely and how long ago. Before the issue day, caught fire burned where the vegetation was greener and denser than in its surroundings (NDVI +0.068, leaf area index +0.24, relative to other land of the same patch) and the soil had dried

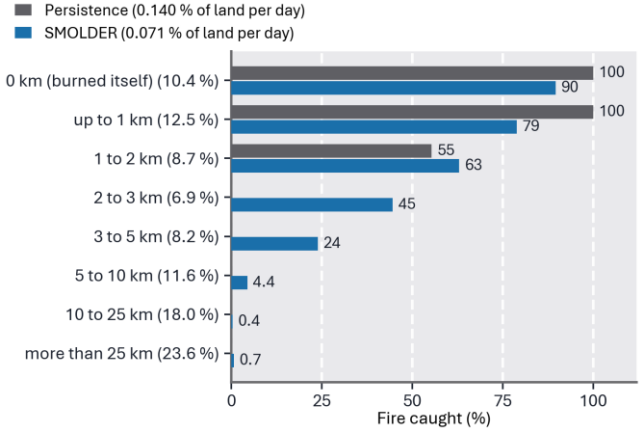


Figure 3. Fire caught in 2020 by distance to the nearest fire detected on the issue day or the two days before ('0 km': the pixel itself burned), for persistence and SMOLDER at their thresholds chosen on 2019 (best F_2). In brackets the share of all 2020 fire in each band. Persistence catches no fire beyond 2 km (no bar).

out, whereas missed fire burned on moister soil with less vegetation (NDVI +0.041, leaf area index +0.13). Air temperature did not distinguish the two.

An upper bound for better weather information was tested on 2019 with a model that also received the actual weather of the three target days, as if the weather forecast were perfect. Its AUC-PR rose only from 0.2062 to 0.2153, and the new fire caught with 0.5 % of the land from 24.8 to 26.4 %. Both runs used an earlier version of the inputs, so only the size of the gain is informative.

5 Discussion and conclusion

5.1 How good is the model?

The absolute numbers look low: in 2020 six of seven flagged pixels did not burn within three days. They have to be read against three facts. First, fire is extremely rare, and the precision of SMOLDER is 431 times the base rate of 2020 (persistence 199 times). Second, much of the fire cannot be predicted from these data by any ranking: of the 41.7 % of the 2020 fire that lay more than 10 km from fire of the days before, SMOLDER caught 0.6 % and persistence none, whereas of the fire within 10 km SMOLDER caught 51.8 % and persistence 47.6 %, with half the flagged area. Third, SMOLDER is better than persistence, the only reference evaluated on the same task, on 348 of 350 days. Scores of other models refer to other tasks: in the WildfireSpreadTS benchmark for next-day fire spread in the United States (Gerard et al., 2023), the best model reaches 1.9 times the average precision of persistence (0.372 against 0.193), but resolution, lead time and evaluated area differ.

The model also has clear weaknesses. Within 3 km of recent fire it catches no more fire than persistence (72.0 % in 2020), and its advantage there is the smaller area. New fire is predicted only weakly (AUC-PR 0.0058 against 0.0022), isolated ignitions not at all, and the catch rates near fire were lower in 2020 than in 2019.

5.2 What the results mean in practice

On an average day of 2020, about 2200 of the 6.9 million land pixels had fire in the following three days. SMOLDER flagged about 4900 pixels (about 5400 km²), of which about 670 burned, 30 % of all the fire. Persistence flagged about 9700, of which about 610 burned. Someone monitoring the flagged pixels would thus watch half the area and see slightly more of the coming fire (Fig. 2). Nearly all of this fire is the spread or re-ignition of fires that are already burning. SMOLDER therefore answers the question ‘where will the fires that are burning now be in the next three days?’, which could help to direct aerial surveillance, crews or warnings around active fires. It does not answer the question ‘where will a new fire start?’, which concerned two fifths of the fire of 2020. For that, fire danger ratings, which describe how dangerous the conditions are wherever a fire might start, and information on ignitions such as lightning remain necessary.

5.3 Why new fires are hard to predict

Fire far from earlier fire is hardly predicted, and the likely reasons are a matter of interpretation. The model has learned that places without recent fire are low risk, which is right most of the time and wrong for exactly these fires. And the cause of an isolated ignition, a lightning strike or a human spark, is a point event that leaves no trace in daily 1 km maps: of the four switches of Sect. 1, the inputs describe fuel, dryness and weather, but not ignition. Neither

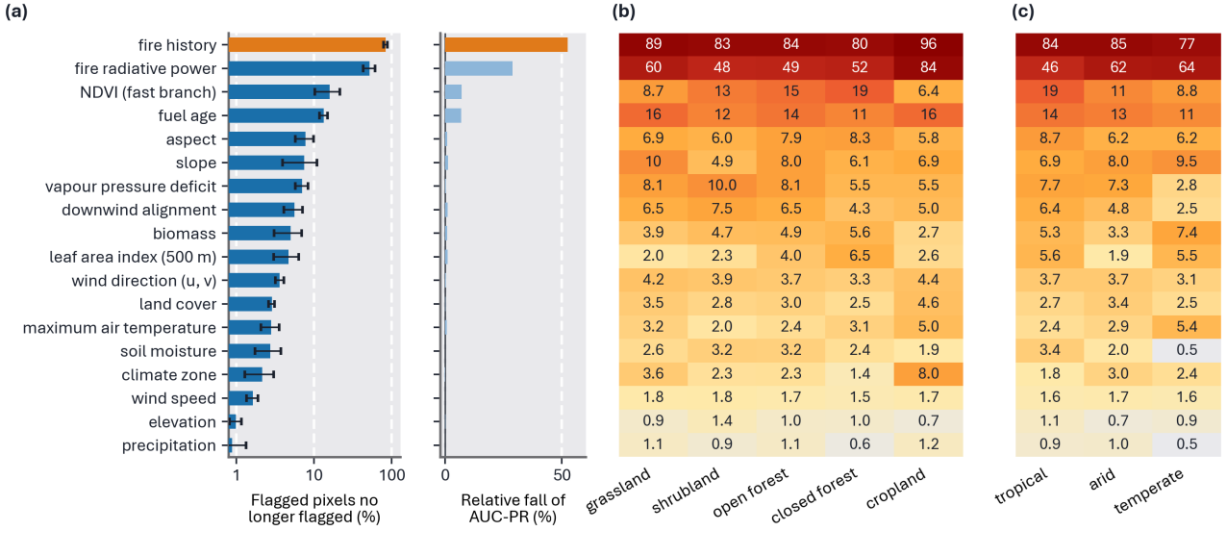


Figure 4. Importance of the inputs, 1500 fire-active patches of 2020. (a) Share of the flagged pixels (threshold chosen on 2019) that are no longer flagged when the input is shuffled (left, with the larger standard deviation over five time blocks and five regions as error bar) and relative fall of AUC-PR (right), (b) by land cover and (c) by climate zone. Longer bars and darker cells mean that the forecast relies more on the input.

the training nor missing weather seems to be the limit: a stronger loss weight for distant fire lowered the validation score (Appendix B), and even the actual weather of the target days added only about 0.009 to the AUC-PR.

Most of the skill of SMOLDER, and probably of any model that uses recent fire as an input, lies within a few kilometres of fire that is already burning. There the fire inputs decide, while weather and fuel at 1 to 4.4 km and one value per day separate neighbouring pixels only weakly. The predictable part of the problem is fire spread and re-ignition near existing fires, at scales of hundreds of metres and hours, below the resolution of this model. It is better framed as a spread model in a window of about 25 km around active fires, as in datasets for next-day fire spread (Huot et al., 2022; Gerard et al., 2023), but with hourly weather and with active fire from geostationary satellites such as Himawari-8/9, which image Australia every 10 minutes at a nominal 2 km in the infrared (Bessho et al., 2016). Where and when new fires start is a different, largely probabilistic problem, for which fire danger ratings and ignition models remain the appropriate tools.

5.4 Lessons for building and evaluating such models

Three points from this study apply to data-driven fire forecasts in general. First, inputs derived from satellite products should be checked for information from after the issue day (Sect. 2.3). Composites, temporal smoothing, gap filling and multi-year maps can all carry the trace of a fire back in time, and this cannot be seen in the scores. A check of each input around the first detection of new fires, as done here, finds strong look-ahead with little effort. Second, a model that receives the current fire as an input should be compared with persistence and evaluated by distance to existing fire. Here the share of fire caught falls from 72.0 % within 3 km of recent fire to 0.6 % beyond 10 km, which a single overall score hides. Third, scores should be reported relative to the base rate and at an operating point fixed on other data. From 2019 to 2020 the base rate fell by 42 % and the AUC-PR of both rankings with it, so absolute scores of years or studies with different amounts of fire cannot be compared directly.

5.5 Limitations and future work

The labels are satellite detections, which miss fires under cloud or dense canopy and small or short-lived fires. Only one test year was evaluated, and it starts while the 2019/20 Black Summer fires were still burning (Boer et al., 2020). The model was not compared on the same task with an operational fire danger index or with other learned models, including simpler ones such as gradient-boosted trees on the same inputs, and the variation between repeated training runs was not quantified. Its output is a ranking and not a probability, and the LAI is at least a month old when it enters a forecast. The most promising routes are a spread model at higher resolution around active fires (Sect. 5.3) and, for new fires, inputs that describe ignition, such as daily lightning data.

5.6 Conclusion

The three questions of the introduction can now be answered. First, it is better than persistence: in the test year 2020 its AUC-PR is 3.3 times that of persistence, and with one threshold chosen on 2019 it catches 30.5 % of the fire on 0.071 % of the land per day, where persistence catches 27.8 % on 0.140 %. Second, its skill comes from fire that is already burning: 91.1 % of the fire it catches lies within 3 km of recent fire, and fires that start far from existing fire are hardly predicted. Third, it relies mainly on the fire record, the fire intensity and the time since an area last burned, whereas weather and fuel add little at 1 km and daily resolution. Models of this kind can anticipate how burning fires develop across a continent, but new fires need information on ignition. Any such model should be judged against persistence, by distance to existing fire and relative to the base rate.

Code and data availability

The code of SMOLDER, the trained model weights and the figures of the supplement are available at https://github.com/kmueller00/smolder-wildfire-australia (MIT licence). The version used in this paper is release v2.0 (tag v2.0, commit e4ab504). The supplement (folder figures/supplement) holds annual risk maps, example forecasts, training curves and maps of the input data. The input data needed to reproduce the evaluation on the test year 2020 (all inputs and the target on the 0.01° grid) are archived at Zenodo, https://doi.org/10.5281/zenodo.23158319, as seven archives and a checksum file (24.09 GB, all versions: https://doi.org/10.5281/zenodo.21749289). They are licensed CC BY-SA 4.0, because the precipitation data are derived from ANUClimate 2.0, which is licensed CC BY-SA 4.0. The years 2015 to 2019 are not archived because of their size and can be rebuilt from the public sources of Table 1 with the scripts in the repository. Each input dataset remains under its provider's licence.

Competing interests

The author declares that there is no conflict of interest.

Acknowledgements

We acknowledge the use of data and imagery from NASA's Fire Information for Resource Management System (FIRMS), part of NASA's Earth Science Data and Information System (ESDIS).

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Appendix A: Model inputs, architecture and training

The two input stacks have the shape (time steps, height, width, channels) $(18, 384, 384, 21)$ for the slow and $(14, 384, 384, 33)$ for the fast encoder. The fast stack contains VPD, maximum air temperature, wind speed, the two wind components, NDVI, the fire history (three binary maps of three days each, the newest ending on the day of the time step, and the proximity to recent fire, $\exp(-d/5\,\mathrm{km})$), three FRP layers, the fuel age and the downwind alignment, the cosine of the angle between the wind and the direction from the nearest fire to the pixel. Both stacks contain six static layers (biomass, land mask, elevation, slope, east and north component of the aspect) and the day of the year as sine and cosine, and land cover and climate zone are encoded as learned embeddings of 6 and 4 channels. Each ConvLSTM encoder (Shi et al., 2015) uses 5×5 kernels and a hidden state of 64 channels. In the fusion, a learned projection of the final slow state is added to the fast state at every step, the sum is normalized (layer normalization) and the slow state appended, giving 128 channels. Formally this is encoder-decoder attention (Vaswani et al., 2017) with the slow state as the only key, so the attention weight is always 1. A 5×5 convolution with a sigmoid turns the 128 channels into a score, and the output of the last fast step is the forecast.

Training samples are patches drawn from the training years, 2000 per epoch, half of them with at least 45 fire pixels in the target. In 30 % of the patches fire must occur where the newest fire map shows none, and in 30 % of the samples all inputs that reveal recent fire are set to the state of no fire. The loss is the binary cross-entropy over land pixels, with fire pixels weighted 100 times at first, lowered to 20 times over the first 8 epochs, isolated fire pixels counted up to three times, softened targets (0.9 and 0.02), a Dice loss (Milletari et al., 2016) in the first three epochs and the same loss on the 13 earlier fast steps with a weight of 0.3. The model was trained for 50 epochs with AdamW and a cosine learning rate from 3×10^{-4} to zero, and the weights of the three epochs with the highest validation score (epochs 20, 34 and 37) were averaged. This is checkpoint averaging as in Vaswani et al. (2017), related to stochastic weight averaging (Izmailov et al., 2018). Of two models trained with and without random flips and rotations of the patches, the one without had the higher AUC-PR on 2019 (0.2091 against 0.2043) and is the final model.

Appendix B: Alternatives that were tested and not adopted

The fire inputs were chosen on 2019 with gradient-boosted trees on pixels within 25 km of recent fire, in which the score of a preliminary version of the model was a feature on both sides of each comparison. Within 3 km of fire, FRP raised the AUC-PR by 13 %, the wind direction relative to the fire with terrain features by 7 % and the fuel age by 3 %, and at 10 to 25 km the fuel age raised it by 34 %.

Four further alternatives were tested on the validation year 2019 and are not part of the final model. Training runs were compared by their best validation AP within the first 10 epochs, for which the reference run reached 0.0898, and all four comparisons were made before the input timing of Sect. 2.3 was final, with the same inputs on both sides. A higher loss weight for fire far from earlier fire, which multiplied the loss of every fire pixel by $1 + a \cdot \min(d/10, 1)$, with d the distance in pixels to the nearest fire of the last 32 days and $a = 5$, 10, 20, 40 and 80, lowered the validation AP to 0.0583 to 0.0736. A standardized anomaly of the VPD as an additional input gave 0.0795, and VPD from the coarser ERA5-based dataset of Montes et al. (2021) instead of BARRA-C2 gave 0.0794. Wind at 05 UTC, the afternoon in Australia, instead of the daily mean brought no gain in AUC-PR within 3 km of fire in the gradient-boosted trees (−0.0010, 95 % interval −0.0077 to +0.0052).

Appendix C: Example events

Figures C1 and C2 show windows of 192 × 192 km, chosen automatically: the four windows in which the F_2 of SMOLDER exceeds that of the baseline the most, and the four in which it is lowest, with at most one window per 30 days and 300 km. The examples are not typical. Of all 1844 fire-active windows of 2020, SMOLDER has the higher F_2 in 73 %, but an F_2 of 0.5 or more only in 15 %. A good forecast shows much dark red (fire inside the flagged area), little orange (fire outside it) and little light blue (flagged land that did not burn).

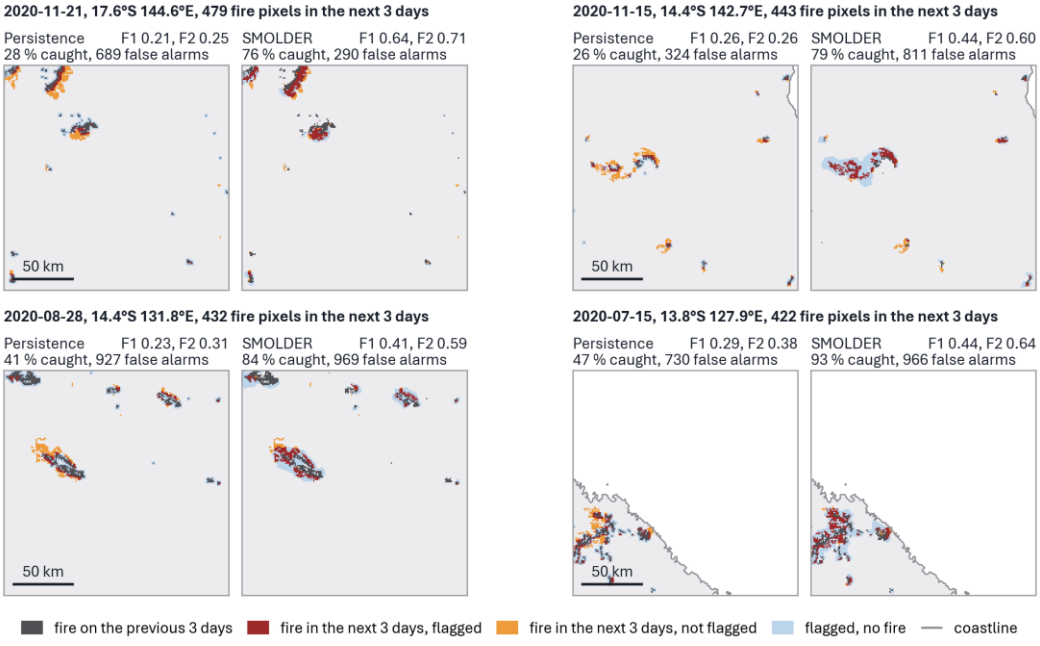


Figure C1. The four fire events of 2020 in which SMOLDER improved most on persistence, each a 192 × 192 km window. Left: persistence, right: SMOLDER, each flagging at its threshold chosen on 2019 (best F_2). Light blue: flagged area without fire. Dark red: fire in the next 3 days inside the flagged area. Orange: fire in the next 3 days outside it. Dark grey: fire on the previous 3 days. Grey line: coastline. Above each event: date of issue, location and the number of fire pixels in the next 3 days. Above each panel: F_1 and F_2 of the ranking within the window, the share of the fire caught and the number of false alarms.

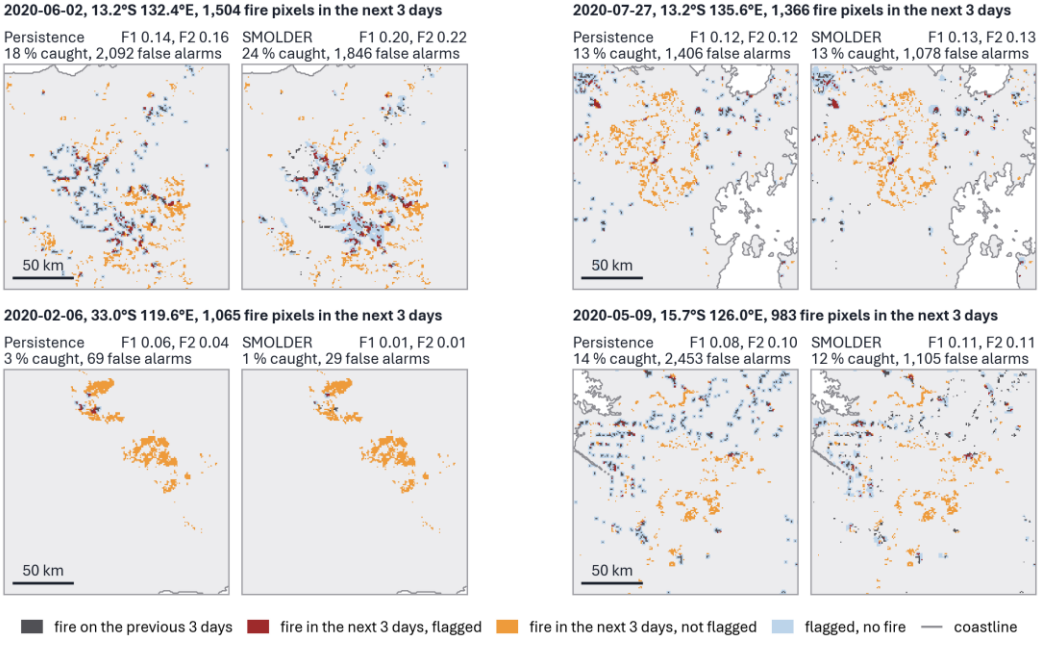


Figure C2. The four fire-active events of 2020 in which SMOLDER did worst, each a 192 × 192 km window. Colours and labels as in Fig. C1.